Skill- 2

Computer Networks

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1 — Install tools

sudo apt update

sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd

```
rohita@ROHITA: ~
rohita@ROHITA:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
118 packages can be upgraded. Run 'apt list --upgradable' to see them.
rohita@ROHITA:~$
```

2 — Build the network

```
tem1
rohita@ROHITA:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
[sudo: authenticate] Password:
rohita@ROHITA:~$  sudo ip netns add host1
Cannot create namespace file "/run/netns/host1": File exists
rohita@ROHITA:~$ sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
rohita@ROHITA:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
rohita@ROHITA:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
rohita@ROHITA:~$
```

3 — Configure IP addresses (TEM 1)

Host

Router

```
rohita@ROHITA:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
rohita@ROHITA:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
rohita@ROHITA:~$
```

Host 2-

```
rohita@ROHITA:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
rohita@ROHITA:~$
```

4 — Enable routing

sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1

sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2

sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1

```
rohita@ROHITA:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
```

5 — Test connectivity

sudo ip netns exec host1 ping -c 3 10.0.2.2

```
rohita@ROHITA:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.559 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.169 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.153 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2042ms
rtt min/avg/max/mdev = 0.153/0.293/0.559/0.187 ms
rohita@ROHITA:~$
```

6 — Add delay and loss

sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss 1%

Verify: sudo ip netns exec router tc qdisc show dev veth-r2

```
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root net
em delay 20ms loss 1%
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 15 limit 1000 delay 20ms loss 1% seed 15412695
719029273207
rohita@ROHITA:~$
```

7 — Open 4 terminals

8 — Start three captures simultaneously

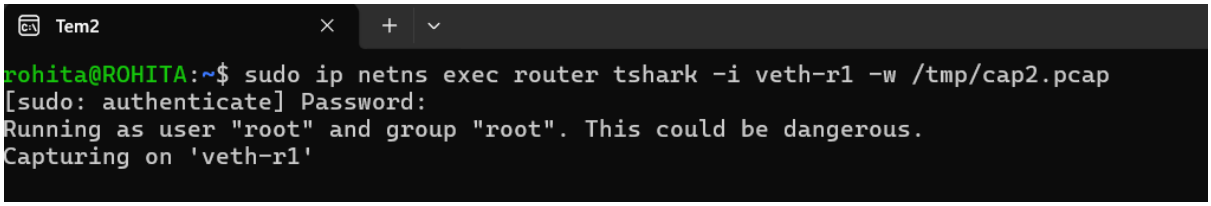
Tem 1:

tem1

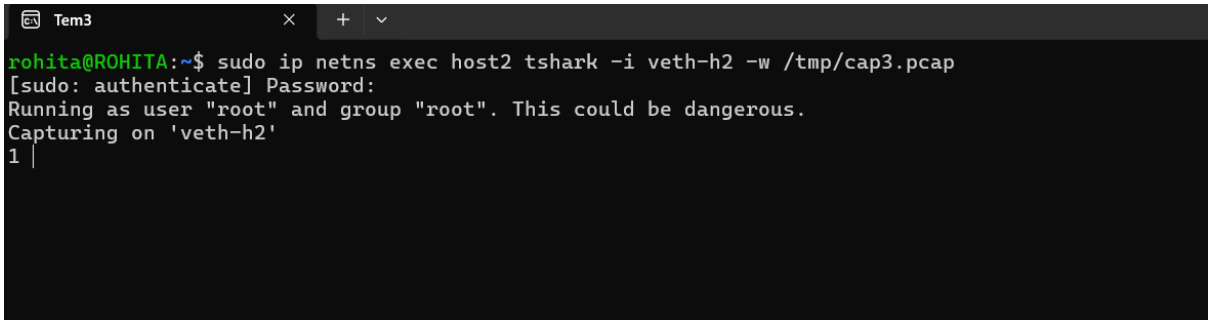
```
rohita@ROHITA:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
rohita@ROHITA:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h
1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
rohita@ROHITA:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-
r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
rohita@ROHITA:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h
2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
rohita@ROHITA:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
rohita@ROHITA:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=0.559 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.169 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.153 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2042ms
rtt min/avg/max/mdev = 0.153/0.293/0.559/0.187 ms
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root net
em delay 20ms loss 1%
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 15 limit 1000 delay 20ms loss 1% seed 15412695
719029273207
rohita@ROHITA:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pca
p
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
1
```

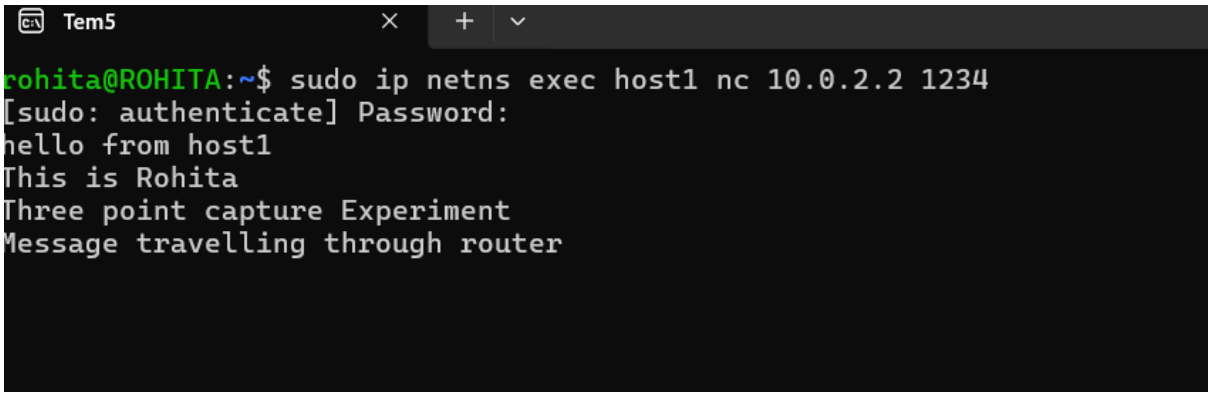
Tem2 :



Tem3:

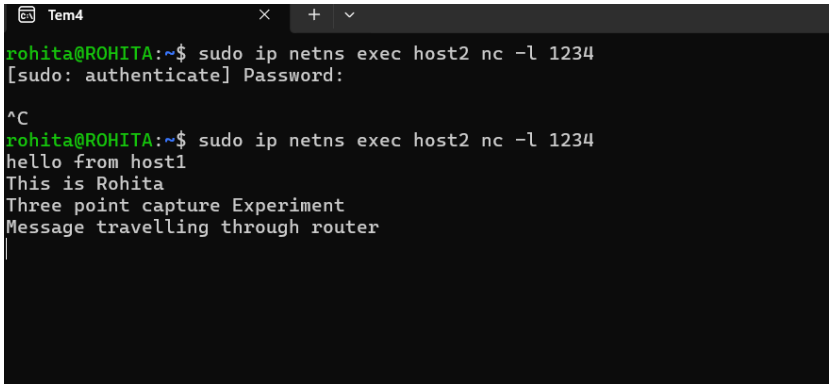


9 — Send real message from host1 to host2



Tem5 – host1

Tem4 – host2



```
rohita@ROHITA:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pca
p
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
20
```

```
Tem2

rohita@ROHITA:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
19
```

```
Tem3

rohita@ROHITA:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
24
```

Packets showing 20,19,24

12 , 13 — Fix permissions , Check TTL

```
rohita@ROHITA:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
[sudo: authenticate] Password:
rohita@ROHITA:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
      5 63
      6 64
      6 63
      5 64
rohita@ROHITA:~$
```

14 — Read your actual message from the capture

```
rohita@ROHITA:~$ sudo apt install -y tshark iproute2 netcat-openbsd wi
[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 118
rohita@ROHITA:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
```

VERIFY

```
rohita@ROHITA:~$ sudo ip netns list
host2 (id: 2)
router (id: 1)
host1 (id: 0)
rohita@ROHITA:~$
```

Create cables

```
rohita@ROHITA:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
rohita@ROHITA:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
rohita@ROHITA:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
rohita@ROHITA:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
Error: ipv4: Address already assigned.
Error: ipv4: Address already assigned.
rohita@ROHITA:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
rohita@ROHITA:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
rohita@ROHITA:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=23.1 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=21.3 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=21.4 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 21.266/21.898/23.071/0.830 ms
rohita@ROHITA:~$
```

3 packets transmitted, 3 received, 0% packet loss

```
 hash [-lr] [-p pathname] [-dt] [name ...]                                    { COMMANDS ; }
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc add veth-r2 root netem delay 20ms loss 1%
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc add veth-r2 root netem delay 20ms loss 1%
Unknown qdisc "veth-r2", hence option "root" is unparsable
rohita@ROHITA:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc netem 8001: root refcnt 15 limit 1000 delay 20ms loss 1% seed 15412695719029273207
rohita@ROHITA:~$
```

Tem 1:

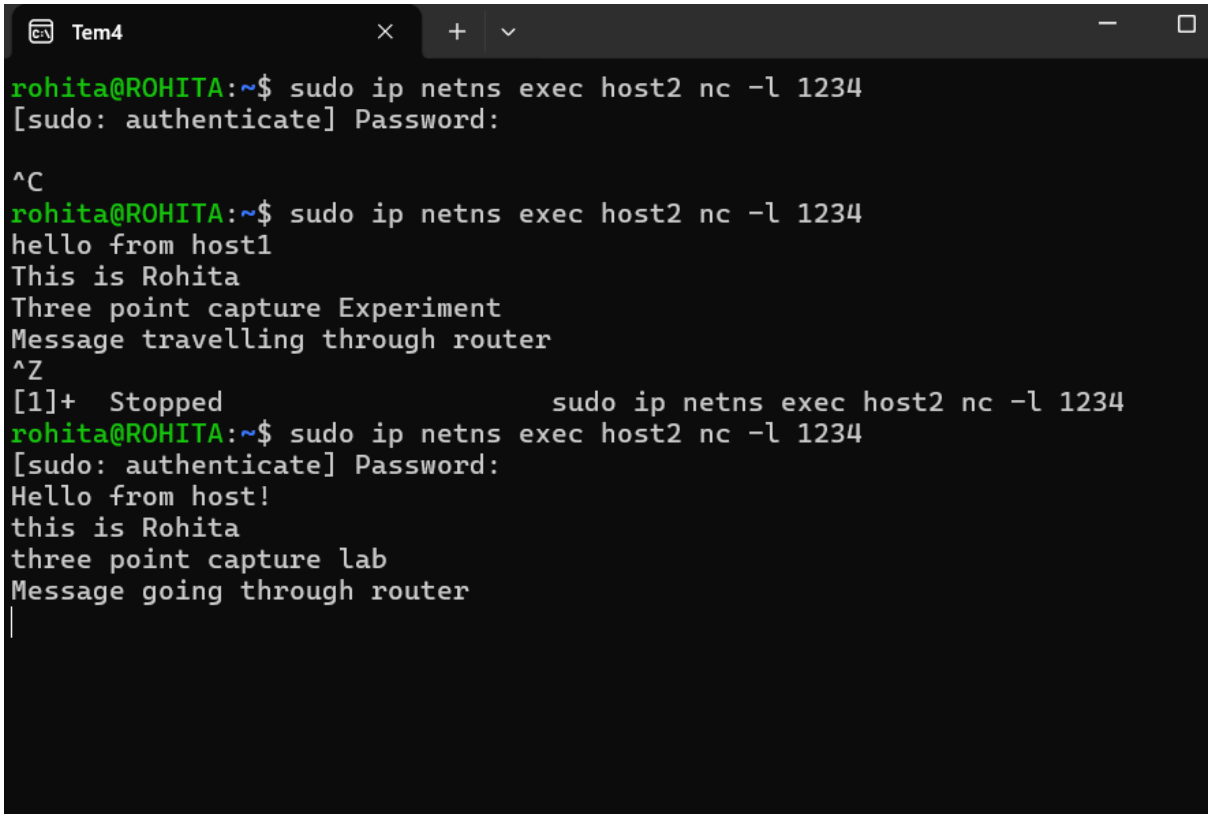
```
qdisc netem 8001: root refcnt 15 limit 1000 delay 20ms loss 1% seed 15412695719029273207
rohita@ROHITA:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
17
```

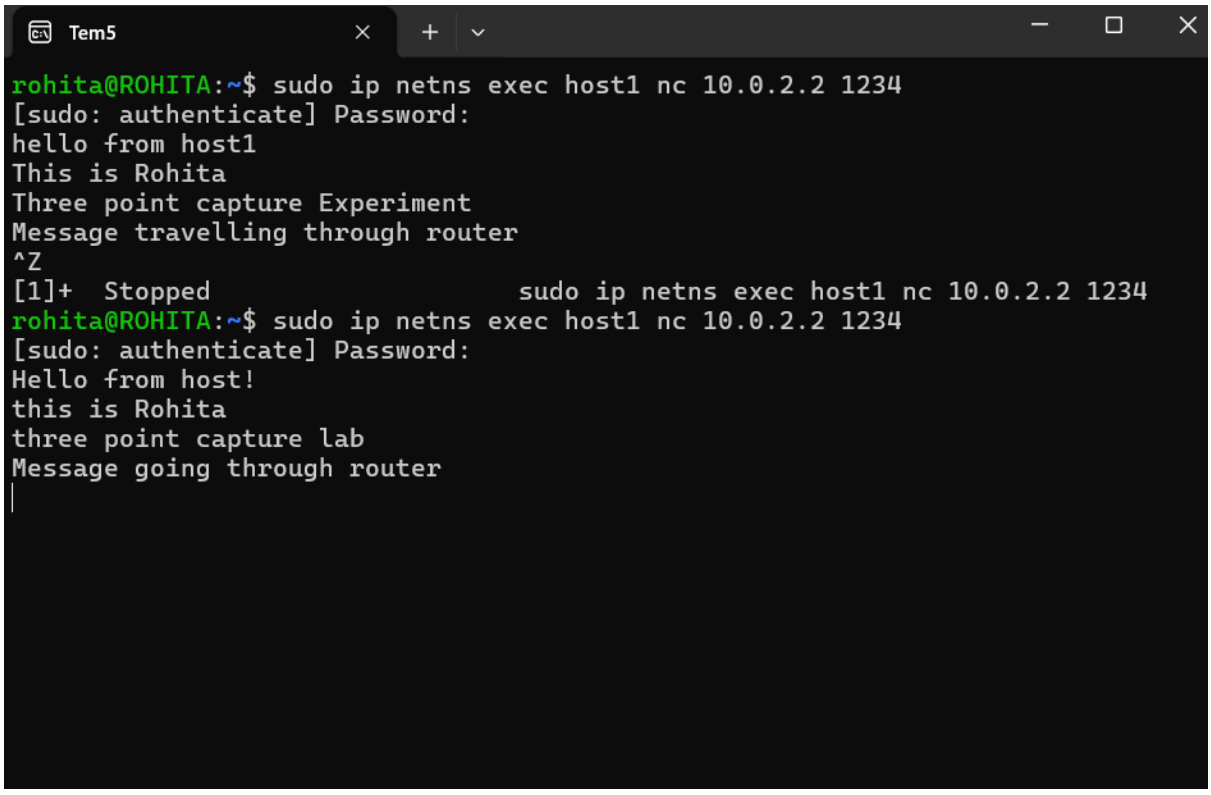
Tem2

```
rohita@ROHITA:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
25 ^C
rohita@ROHITA:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
17
```

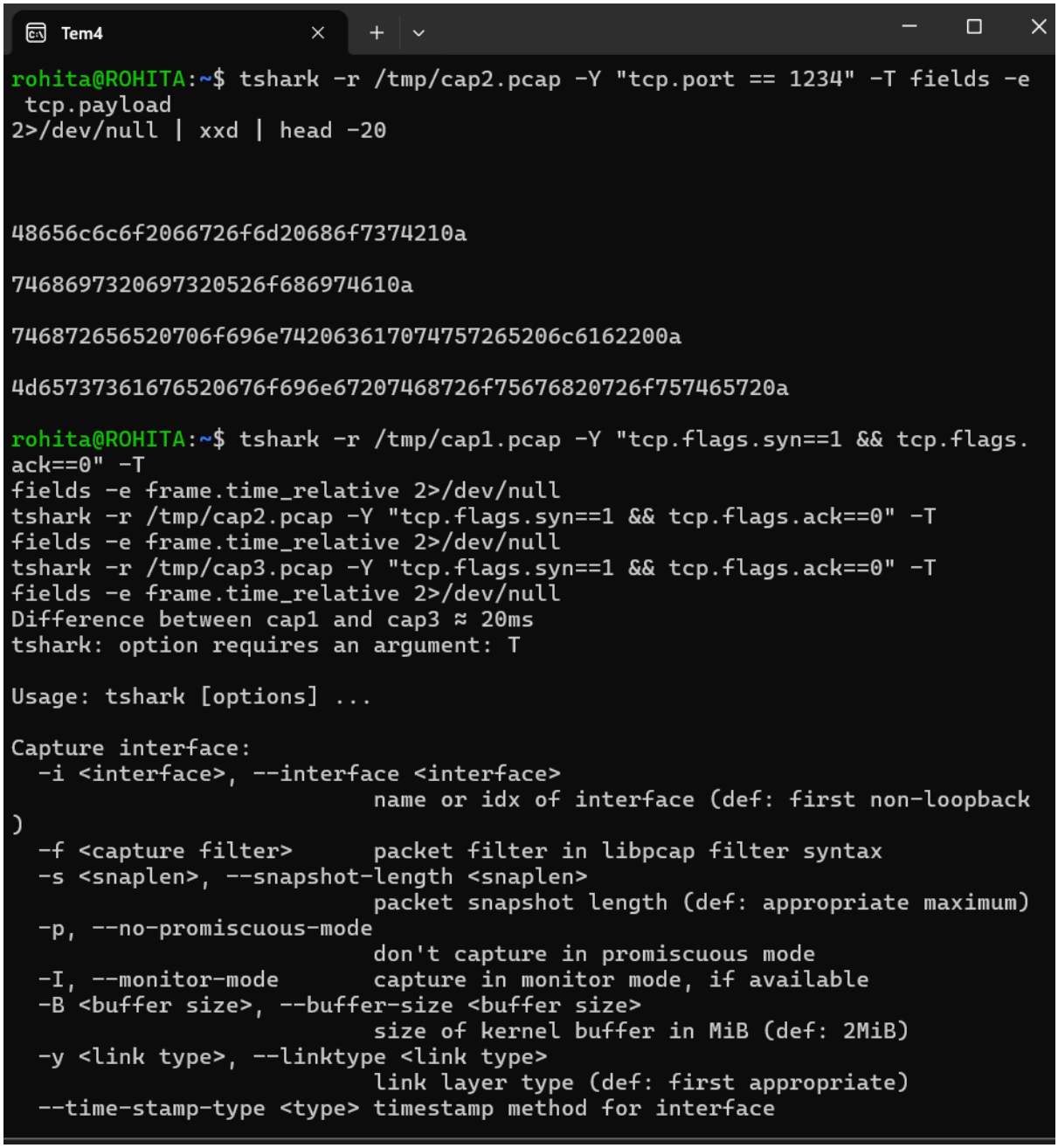
Tem3

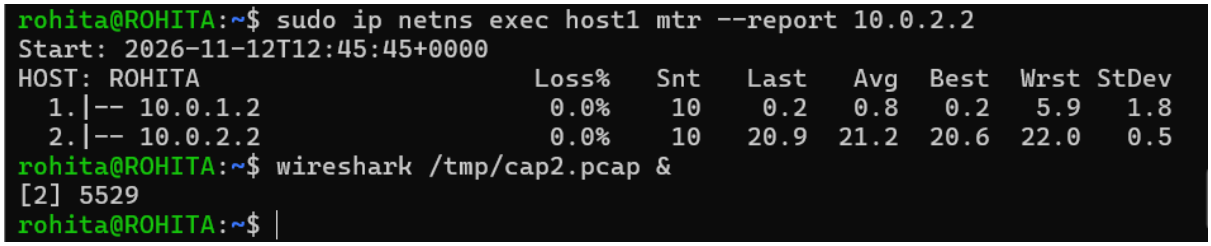
```
rohita@ROHITA:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
29 ^Z
[1]+  Stopped                 sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
rohita@ROHITA:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
15
```

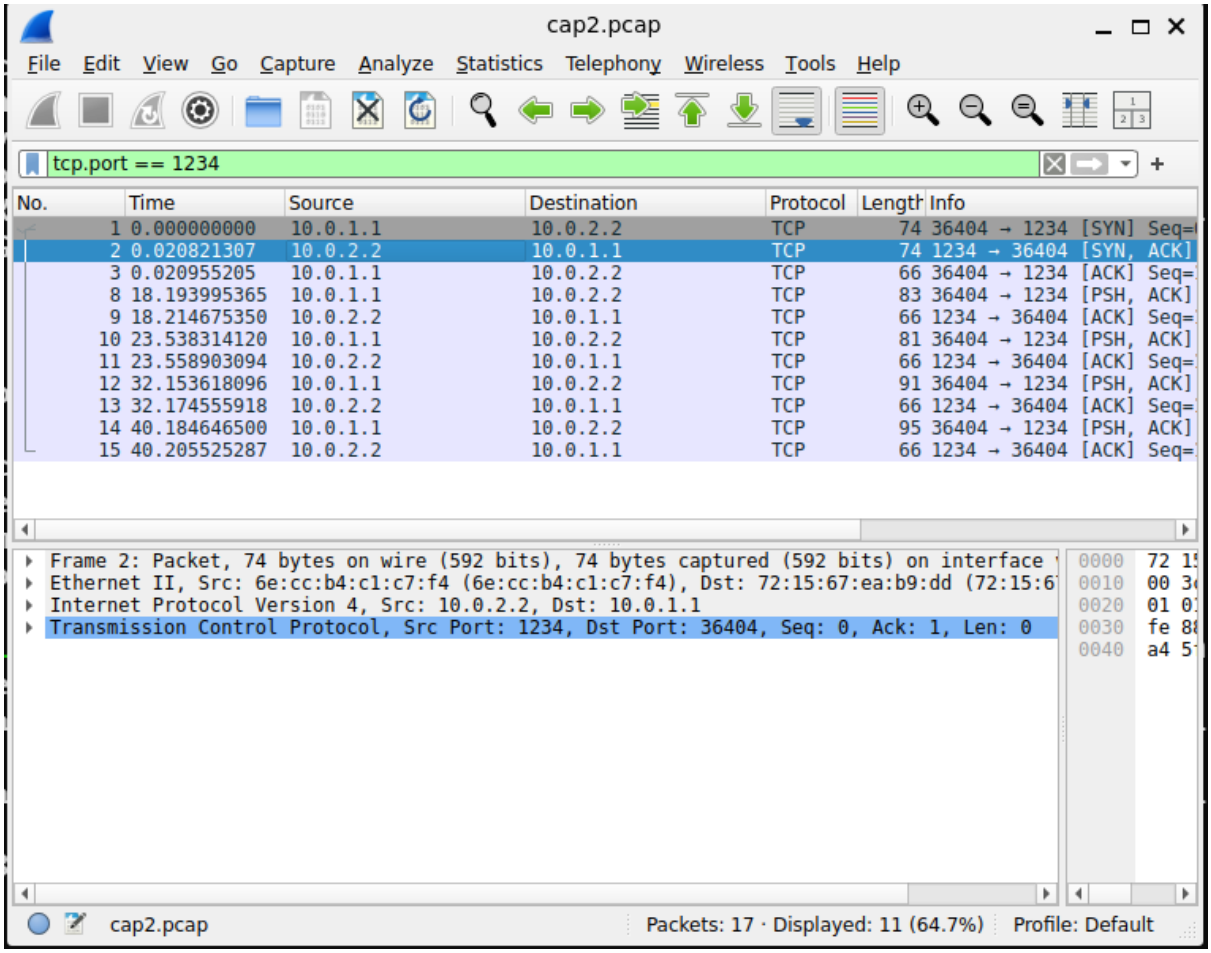

Tem4
rohita@ROHITA:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^C
rohita@ROHITA:~$ sudo ip netns exec host2 nc -l 1234
hello from host1
This is Rohita
Three point capture Experiment
Message travelling through router
^Z
[1]+ Stopped sudo ip netns exec host2 nc -l 1234
rohita@ROHITA:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
Hello from host!
this is Rohita
three point capture lab
Message going through router


Tem5
rohita@ROHITA:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
hello from host1
This is Rohita
Three point capture Experiment
Message travelling through router
^Z
[1]+ Stopped sudo ip netns exec host1 nc 10.0.2.2 1234
rohita@ROHITA:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host!
this is Rohita
three point capture lab
Message going through router

```
rohita@ROHITA:~$ sudo chmod 644 /tmp/capl.pcap /tmp/cap2.pcap /tmp/cap3.pcap
chmod: cannot access '/tmp/capl.pcap': No such file or directory
rohita@ROHITA:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 2.0K Nov 12 12:41 /tmp/cap1.pcap
-rw-r--r-- 1 root root 2.0K Nov 12 12:41 /tmp/cap2.pcap
-rw-r--r-- 1 root root 1.9K Nov 12 12:41 /tmp/cap3.pcap
rohita@ROHITA:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev
/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | un
iq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | un
iq -c
      5 63
      6 64
      5 63
      6 64
      6 63
      5 64
rohita@ROHITA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e
 tcp.payload
2>/dev/null | xxd | head -20
```


Tem4
rohita@ROHITA:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e
tcp.payload
2>/dev/null | xxd | head -20
48656c6c6f2066726f6d20686f7374210a
7468697320697320526f686974610a
7468726565520706f696e74206361707475726520 6c6162200a
4d65737361676520676f696e67207468726f756768207265746f726d20a
rohita@ROHITA:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.
ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
Difference between cap1 and cap3 ≈ 20ms
tshark: option requires an argument: T
Usage: tshark [options] ...
Capture interface:
-i <interface>, --interface <interface>
name or idx of interface (def: first non-loopback
)
-f <capture filter> packet filter in libpcap filter syntax
-s <snaplen>, --snapshot-length <snaplen>
packet snapshot length (def: appropriate maximum)
-p, --no-promiscuous-mode
don't capture in promiscuous mode
-I, --monitor-mode capture in monitor mode, if available
-B <buffer size>, --buffer-size <buffer size>
size of kernel buffer in MiB (def: 2MiB)
-y <link type>, --linktype <link type>
link layer type (def: first appropriate)
--time-stamp-type <type> timestamp method for interface


rohita@ROHITA:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T12:45:45+0000
HOST: ROHITA Loss% Snt Last Avg Best Wrst StDev
1.|-- 10.0.1.2 0.0% 10 0.2 0.8 0.2 5.9 1.8
2.|-- 10.0.2.2 0.0% 10 20.9 21.2 20.6 22.0 0.5
rohita@ROHITA:~$ wireshark /tmp/cap2.pcap &
[2] 5529
rohita@ROHITA:~$




```
rohita@ROHITA:~$ sudo ip netns del host1
sudo ip netns del router
sudo ip netns del host2
[sudo: authenticate] Password:
rohita@ROHITA:~$ sudo ip netns list
rohita@ROHITA:~$
```

Everthing is cleared in tem 4 .